

</> CodeWithHarry

PYTHON

CHEATSHEET



```
# Hello World
print("Hello, World!")

# Variables
x = 10          # int
y = 3.14        # float
name = "Harry" # str
is_active = True # bool

# If-Else
if x > 5:
    print("x is greater than 5")
elif x == 5:
    print("x is equal to 5")
else:
    print("x is less than 5")

# For Loop
for i in range(5):
    print(i)

# Function
def greet(name):
    return f"Hello, {name}!"
print(greet(name))
```



BASIC SYNTAX

Indentation, Statements, Comments, Input/Output



DATA TYPES

Numbers, String, List, Tuple, Set, Dictionary



OPERATORS

Arithmetic, Comparison, Logical, Assignment



CONTROL FLOW

if, elif, else, for, while, break, continue



FUNCTIONS

def, return, *args, **kwargs, lambda



COLLECTIONS

List, Tuple, Set, Dictionary



MODULES & PACKAGES

import, from...import, as, pip



EXCEPTION HANDLING

try, except, else, finally, raise



BASIC
SYNTAX



DATA
TYPES



CONTROL
FLOW



FUNCTIONS



MODULES



EXCEPTIONS



FILE
HANDLING

Python Programming

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Introduction & Installation

Installation

WEB

Go to python.org/downloads and install the latest version

Check Python Version

```
python --version
python3 --version
py --version
```

Run Python

```
python main.py
```

Basic Hello World Syntax

PYTHON FILE

```
print("Hello, World!")
```



Python Programming

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Variables & Data Types

Variables

STRING

```
name = "Harry"
```

INTEGER

```
age = 25
```

FLOAT

```
price = 99.99
```

BOOLEAN

```
is_active = True
```

MULTIPLE

```
x, y = 10, 20
```

SAME VALUE

```
a = b = c = 0
```

Data Types

TEXT

```
str
```

NUMBER

```
int, float, complex
```

SEQUENCE

```
list, tuple, range
```

MAPPING

```
dict
```

SET

```
set, frozenset
```

BOOLEAN

```
bool
```

Type Checking

PYTHON

```
x = 5
print(type(x))
print(isinstance(x, int))
```

Constant Naming Convention

Constant names are typically written in uppercase letters. This is a convention, not enforced!

```
PI = 3.14
```

```
MAX_SIZE = 100
```



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Operators & Type Conversion

Operators

ARITHMETIC

+, -, *, /, %, //, **

ASSIGNMENT

=, +=, -=, *=, /=, %=

COMPARISON

==, !=, >, <, >=, <=

LOGICAL

and, or, not

MEMBERSHIP

in, not in

IDENTITY

is, is not

Operator Precedence

1. PARENTHESES

() first

2. POWER

**

3. MULTIPLY

*, /, //, %

4. ADD

+, -

5. COMPARE

==, !=, <, >, in, is

6. NOT

not

7. AND

and

8. OR

or

Operator Syntax

ADD

x + y

FLOOR DIV

x // y

POWER

x ** y

EQUAL

x == y

LOGIC

x > 10 and x < 20

MEMBER

"a" in "harry"

IDENTITY

x is None

Type Conversion

INTEGER (immutable)

int("10")

FLOAT (immutable)

float("10.5")

STRING (immutable)

str(100)

BOOLEAN (immutable)

bool(0) is False;
bool("False") is True

LIST (mutable)

list("abc")

TUPLE (immutable)

tuple([1, 2])

Python Programming

Strings

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Create Strings

SINGLE

```
s = 'hello'
```

DOUBLE

```
s = "hello"
```

MULTILINE

```
s = '''hello'''
```

RAW

```
r"C:\new\file"
```

Indexing & Slicing

Assume s = "Harry"

FIRST

```
s[0] -> "H"
```

LAST

```
s[-1] -> "y"
```

SLICE

```
s[1:4] -> "arr"
```

STEP

```
s[::2] -> "Hry"
```

REVERSE

```
s[::-1] -> "yrRaH"
```

LENGTH

```
len(s) -> 5
```

String Methods

Assume s = "Harry"

UPPER

```
s.upper() ->  
"HARRY"
```

LOWER

```
s.lower() ->  
"harry"
```

SPACES

```
s.strip() ->  
"Harry"
```

REPLACE

```
s.replace("a",  
"o") -> "Horry"
```

SPLIT

```
s.split("r") ->  
["Ha", "", "y"]
```

JOIN

```
"-".join(s) ->  
"H-a-r-r-y"
```

FIND

```
s.find("r") -> 2
```

Formatting

PYTHON

```
name = "Harry"  
age = 25  
print(f"{name} is {age}")
```

Python Programming

Lists

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List Syntax

CREATE

```
lst = [1, 2, 3]
```

MIXED

```
lst = [1, "a", True]
```

EMPTY LIST

```
lst = []
```

LENGTH

```
len(lst)
```

Access & Slice

FIRST/LAST

```
lst[0] / lst[-1]
```

RANGE

```
lst[1:4]
```

REVERSE

```
lst[::-1]
```

List Methods

ADD ONE

```
lst.append(x) adds one  
element
```

ADD MANY

```
lst.extend(items)
```

INSERT

```
lst.insert(i, x)
```

REMOVE

```
lst.remove(x) removes  
first match
```

POP

```
lst.pop()
```

CLEAR

```
lst.clear()
```

SORT

```
lst.sort()
```

REVERSE

```
lst.reverse()
```

Comprehension

PYTHON

```
squares = [x*x for x in range(5)]  
evens = [x for x in nums if x % 2 == 0]
```



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Tuples, Sets & Dictionaries

Tuples

CREATE

```
t = (1, 2, 3)
```

ONE ITEM

```
t = (1,)
```

ACCESS

```
t[0], t[-1]
```

UNPACK

```
a, b = (1, 2)
```

Sets

CREATE

```
s = {1, 2, 3}
```

EMPTY SET

```
s = set()
```

ADD

```
s.add(4)
```

REMOVE

```
s.remove(2) # error if  
not present  
s.discard(2) # safer
```

UNION

```
a | b
```

INTERSECTION

```
a & b
```

Dictionaries

CREATE

```
d = {"name": "Harry"}
```

ACCESS

```
d["name"]
```

GET

```
d.get("age")
```

UPDATE

```
d["age"] = 25
```

KEYS

```
d.keys()
```

ITEMS

```
d.items()
```



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Conditional Statements

if / elif / else

PYTHON

```
if condition:
    statement
elif another_condition:
    statement
else:
    statement
```

Comparisons

EQUAL/ NOT EQUAL

```
x == y
x != y
```

GREATER/LESS

```
x > y
x < y
```

GTE/LTE

```
x >= y
x <= y
```

Logical Conditions

AND/OR

```
x > 0 and x < 10
x == 0 or x == 1
```

NOT

```
not is_active
```

TERNARY

```
a if condition else b
```

Match Case

PYTHON 3.10+

```
match value:
    case 1:
        print("one")
    case _:
        print("other")
```



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Loops

for Loop

PSEUDO CODE

```
for item in iterable:  
    statement
```

PYTHON

```
for i in range(5):  
    print(i)
```

while Loop

PSEUDO CODE

```
while condition:  
    statement
```

PYTHON

```
i = 0  
while i < 5:  
    i += 1
```

Loop Tools

RANGE

```
range(start, stop, step)
```

BREAK

```
break
```

CONTINUE

```
continue
```

PASS

```
pass
```

ENUMERATE

```
enumerate(items)
```

ZIP

```
zip(a, b)
```

Loop else

PYTHON

```
for x in nums:  
    if x == target:  
        break  
else:  
    print("not found")
```

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Functions

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Function Syntax

PYTHON

```
def function_name(parameters):  
    return value  
  
result = function_name(arguments)
```

Parameters

NORMAL

```
def add(a, b):
```

DEFAULT

```
def greet(name="User"):
```

KEYWORD

```
greet(name="Harry")
```

ARGS

```
def f(*args):
```

KWARGS

```
def f(**kwargs):
```

ANNOTATION

```
def add(a: int) -> int:
```

Lambda

PYTHON

```
square = lambda x: x * x  
print(square(5))
```

Scope

LOCAL

x inside function

GLOBAL

global x



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Exception Handling

try / except

PYTHON

```
try:
    risky_code()
except Exception as e:
    print(e)
```

Full Block

PYTHON

```
try:
    code
except ValueError as e:
    print(e)
else:
    print("no error")
finally:
    print("always runs")
```

Common Exceptions

VALUE

ValueError

TYPE

TypeError

INDEX

IndexError

KEY

KeyError

FILE

FileNotFoundError

ZERO DIV

ZeroDivisionError

Raise

PYTHON

```
if age < 0:
    raise ValueError("Invalid age")
```



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File Handling

Open File

READ

```
open("file.txt", "r")
```

WRITE

```
open("file.txt", "w")
```

APPEND

```
open("file.txt", "a")
```

BINARY

```
open("file.bin", "rb")
```

with Syntax

PYTHON

```
with open("data.txt", "r") as f:  
    content = f.read()
```

Read / Write

READ ALL

```
f.read()
```

READ LINE

```
f.readline()
```

READ LINES

```
f.readlines()
```

WRITE

```
f.write("hello")
```



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Object Oriented Programming

Class Syntax

PYTHON

```
class Person:
    def __init__(self, name):
        self.name = name

    def greet(self):
        print(self.name)
```

Object Syntax

CREATE

p = Person("Harry")

ATTRIBUTE

p.name

METHOD

p.greet()

Inheritance

PYTHON

```
class Student(Person):
    def __init__(self, name, marks):
        super().__init__(name)
        self.marks = marks
```

Special Methods

INIT

__init__(self)

STR

__str__(self)

LEN

__len__(self)

REPR

__repr__(self)



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Modules & pip

Import Syntax

MODULE

```
import math
```

ALIAS

```
import numpy as  
np
```

FROM

```
from math import  
sqrt
```

ALL

```
from module  
import *
```

Module Usage

PYTHON

```
import math  
print(math.sqrt(16))  
  
from random import randint  
print(randint(1, 10))
```

pip Commands

INSTALL

```
pip install package
```

UNINSTALL

```
pip uninstall package
```

LIST

```
pip list
```

FREEZE

```
pip freeze
```

UPGRADE

```
pip install -U package
```

REQUIREMENTS

```
pip install -r  
requirements.txt
```

Virtual Environment

TERMINAL

```
python -m venv .venv  
.venv\Scripts\activate  
source .venv/bin/activate
```

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Useful Built-in Functions

Data Functions

TYPE

`type(x)`

LENGTH

`len(x)`

INPUT

`input("Name: ")`

OUTPUT

`print(x)`

CONVERT

`int(), float(), str()`

BOOLEAN

`bool(x)`

Math Functions

SUM

`sum(nums)`

MIN

`min(nums)`

MAX

`max(nums)`

ROUND

`round(x, 2)`

ABS

`abs(x)`

POWER

`pow(x, y)`

Iterable Functions

RANGE

`range(5)`

ENUMERATE

`enumerate(items)`

ZIP

`zip(a, b)`

SORTED

`sorted(items)`

ANY

`any(values)`

ALL

`all(values)`

Examples

PYTHON

```
nums = [3, 1, 2]
print(len(nums))
print(sorted(nums))
print(sum(nums))
```


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Useful Built-in Functions

USEFUL BUILT-IN FUNCTIONS

Python provides many built-in functions for common tasks.

print()

Display output

len()

Get length

type()

Get type

range()

Generate sequence

sum()

Sum items

min()/max()

Find min/max

sorted()

Sort items

enumerate()

Index with items

zip()

Combine iterables

map()

Apply function

filter()

Filter items

any()/all()

Test conditions

```
nums = [1, 2, 3]
print(sum(nums))
print(list(enumerate(nums)))
print(list(map(str, nums)))
```

